Cloud Computing Tutorial -4 & 5

CloudSim Simulation

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CLOUDSIM SIMULATION.

Cloudsim is a simulation toolkit used to model & test cloud computing environments without needing real cloud infrastructure.

1. Install Java.
   → verify with "java --version"

2. Download Cloudsim.
   → Go to offical Cloudsim repo.
   → Download .zip file
   → Extract in file system

3. Set up & IDE
   → IntelliJ IDEA

4. Create a Java project
   → Create a new Java project
   → Add Cloudsim .jar files.

5. Import Cloudsim packages.
   → import org.cloudbus.cloudsim.*;

6. Initialize Cloudsim.
   → Cloudsim.init(numUsers, calendar, traceFlag);

7. Create Datacent
   → Define hosts (CPU, RAM, storage
   → Wrap them into a datacenter.

3. Create Broker.
→ DatacenterBroker broker = new DatacenterBroker("Broker");

4. Create VM
→ Define RAM, CPU, bandwidth, storage

5. Create Cloudlets
→ Define length, filesize, outputsize

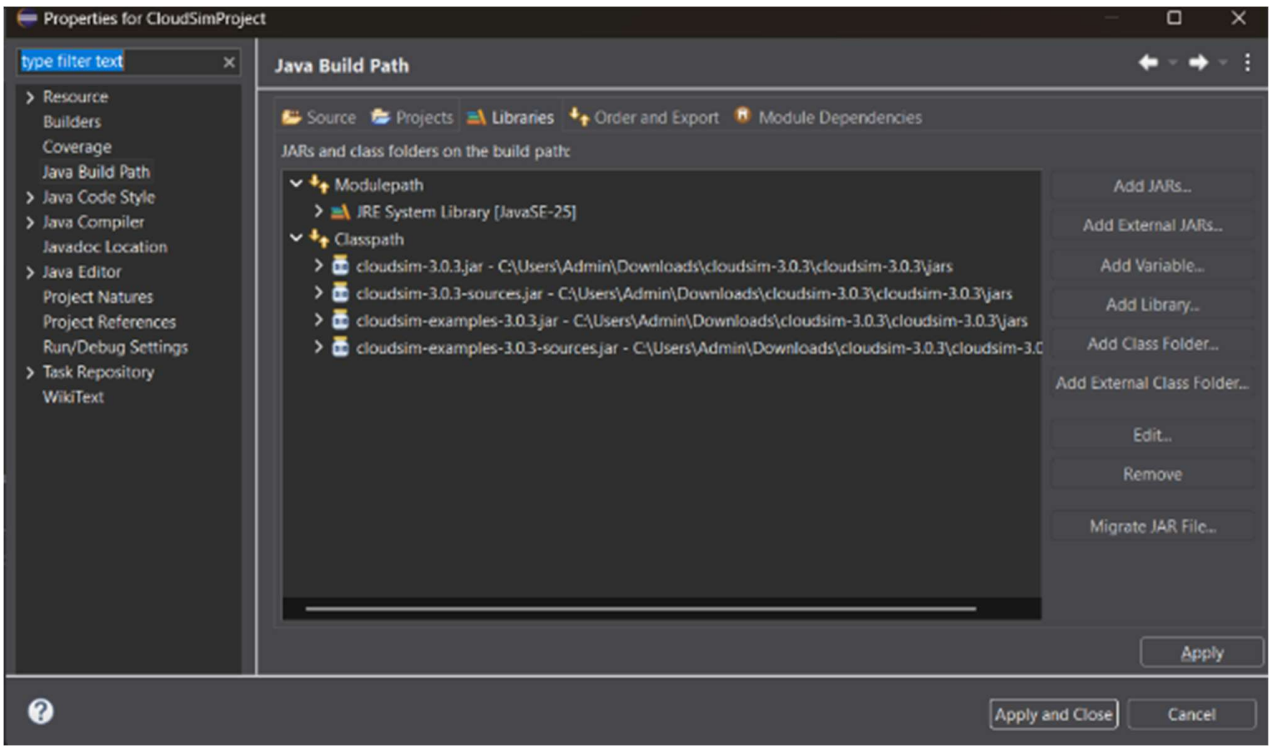
6. Submit VMs & cloudlets to Broker
→ broker.submitVmList(vmList);
→ broker.submitCloudletList(cloudletList);

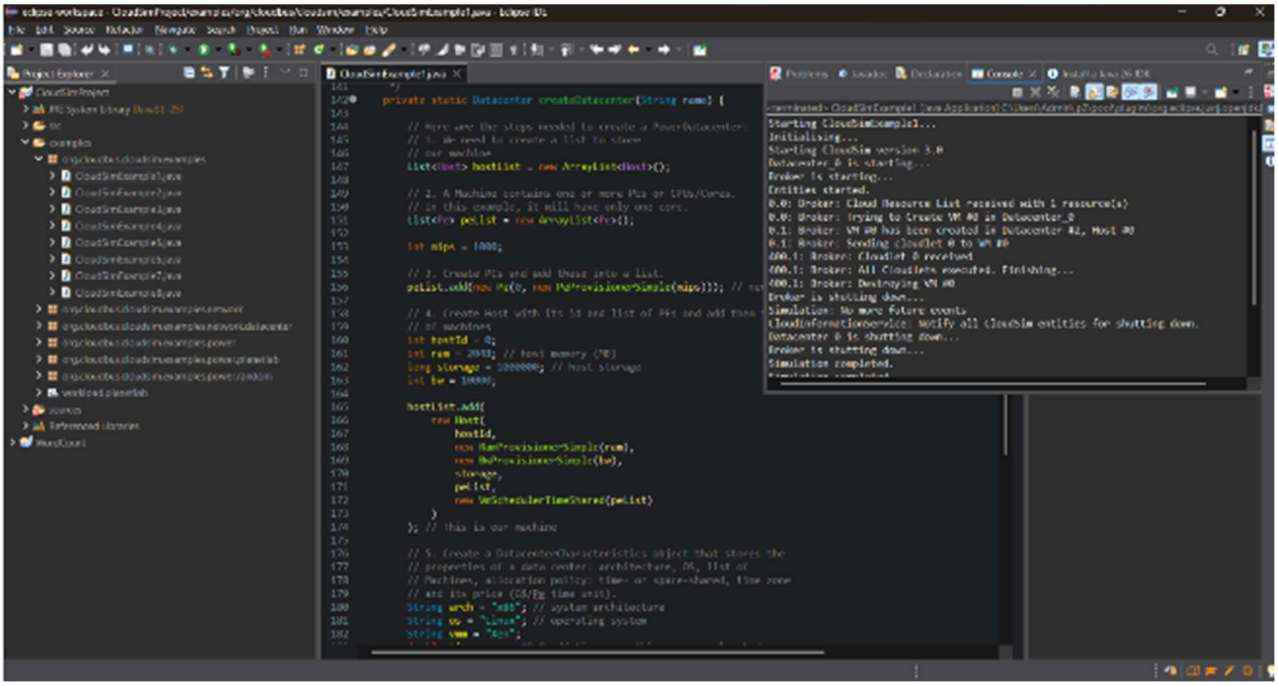
7. Start Simulation.
→ CloudSim.startSimulation();
→ CloudSim.stopSimulation();

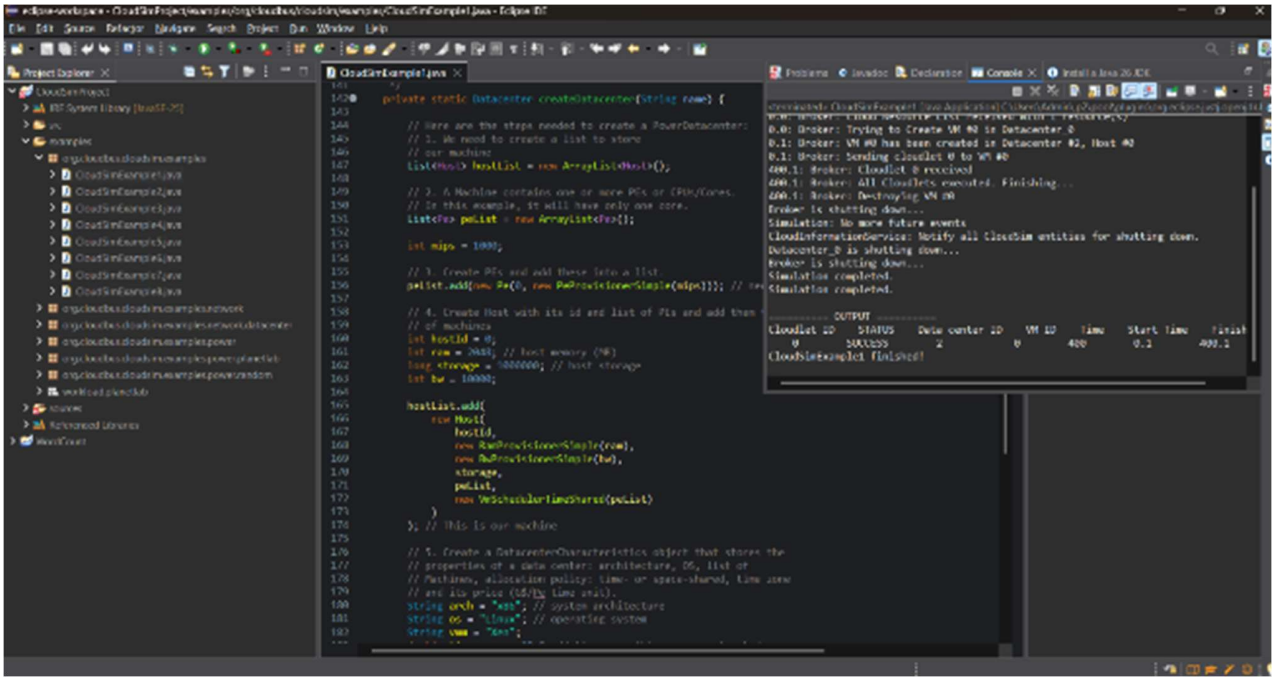
8. Get results.
→ Check execution time, completion status, resource usage.

```
C:\Users\Admin>java --version
java 23.0.1 2024-10-15
Java(TM) SE Runtime Environment (build 23.0.1+11-39)
Java HotSpot(TM) 64-Bit Server VM (build 23.0.1+11-39, mixed mode, sharing)

C:\Users\Admin>
```




Project Explorer
CloudSimExample1.java
private static Datacenter createDatacenter(String name) {
// Here are the steps needed to create a PowerDatacenter:
// 1. We need to create a list to store
// our machine
List<Host> hostList = new ArrayList<Host>();
// 2. A Machine contains one or more PEs or CPUs/Cores.
// In this example, it will have only one core.
List<Pe> peList = new ArrayList<Pe>();
int mips = 1000;
// 3. Create PEs and add these into a list.
peList.add(new Pe(0, new PeProvisionerSimple(mips)));
// 4. Create Host with its id and list of PEs and add them
// of machines
int hostId = 0;
int ram = 2048; // host memory (MB)
long storage = 1000000; // host storage
int bw = 10000;
hostList.add(
new Host(
hostId,
new RamProvisionerSimple(ram),
new BwProvisionerSimple(bw),
storage,
peList,
new VmSchedulerTimeShared(peList)
)
); // This is our machine
// 5. Create a DatacenterCharacteristics object that stores the
// properties of a data center: architecture, OS, list of
// Machines, allocation policy: time- or space-shared, time zone
// and its price (G$/Pe time unit).
String arch = "x86"; // system architecture
String os = "Linux"; // operating system
String vmm = "Xen";
Problems
Javadoc
Declaration
Console
0.0: Broker: Trying to Create VM #0 in Datacenter 0
0.1: Broker: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker: Sending Cloudlet 0 to VM #0
400.1: Broker: Cloudlet 0 received
400.1: Broker: All Cloudlets executed. Finishing...
400.1: Broker: Destroying VM #0
Broker is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_0 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.
========== OUTPUT ==========
Cloudlet ID STATUS Data center ID VM ID Time Start Time Finish
0 SUCCESS 2 0 400 0.1 400.1
CloudletExample1 finished!